

Graphs of Sine and Cosine Functions

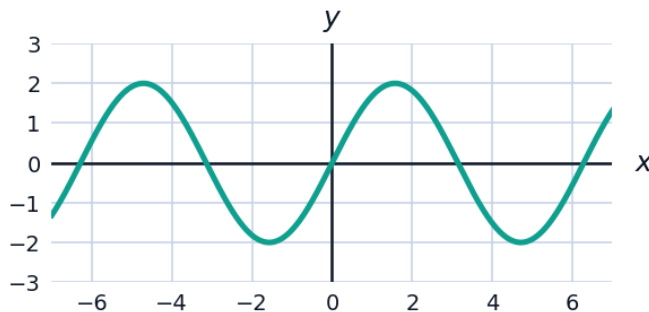


Amplitude, period, and midline

- 1 For the function $y = 3\sin(2x)$, state:
 - a the amplitude
 - b the period
 - c the range
 - 2 For the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:
 - a the amplitude
 - b the period
 - c the midline
 - d the range
 - 3 For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.
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Reading and writing equations from graphs

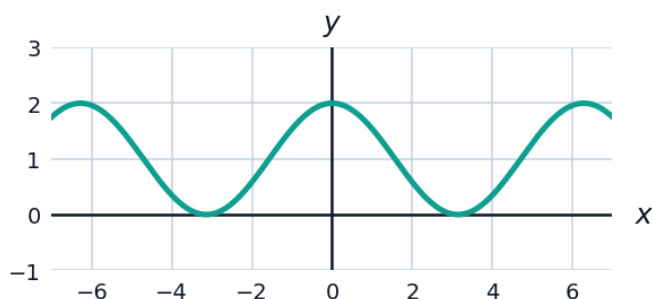
- 4 The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.



- 5 Write an equation of a cosine function with amplitude 4, period π , reflected across the x -axis, with no phase or vertical shift.
 - 6 The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where t is in hours after midnight.
 - a State the maximum and minimum depths.
 - b State the period of the tide cycle.
 - c Find the depth at $t = 3$.
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Graphing with transformations

- 7 The graph below shows $y = \cos(x) + 1$, a cosine curve with a vertical shift. Identify the midline and the maximum and minimum values shown.



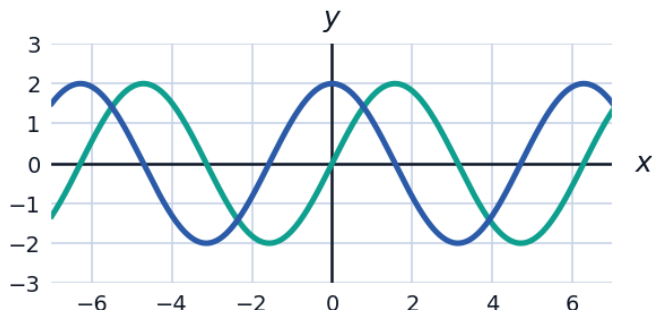
- 8 Describe, in order, the transformations that turn $y = \sin x$ into $y = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.
- 9 State the amplitude, period, phase shift, and midline for $y = 5\cos\left(3x + \frac{\pi}{2}\right) - 4$. (Hint: factor the argument as $3\left(x + \frac{\pi}{6}\right)$.)

Modeling periodic phenomena

- 10 The average monthly temperature in a city varies between 8°C (in January, month 1) and 28°C (in July, month 7), following a sinusoidal pattern with a 12-month period.
- a Find the amplitude and midline of the model.
 - b Write a cosine model $T(m)$ for the temperature in month m , using the fact that the minimum occurs at $m = 1$.
 - c Use your model to estimate the temperature in month 4 (April).

Comparing sine and cosine graphs

- 11 Explain why $\cos x = \sin\left(x + \frac{\pi}{2}\right)$ by comparing the graphs of sine and cosine.
- 12 Both $y = 2\sin(x)$ and $y = 2\cos(x)$ are graphed below. State one key visual difference and one similarity between them.



Finding zeros and extrema from an equation

- 13 Find all zeros of $y = \sin(2x)$ on the interval $[0, 2\pi]$.
- 14 Find the x -value(s) in $[0, 2\pi]$ where $y = 3\cos(x)$ attains its maximum and minimum values.

Solving basic sinusoidal equations

- 15 Solve $2\sin(x) + 1 = 0$ for $x \in [0, 2\pi]$.
- 16 A Ferris wheel's height above ground is $h(t) = 15\sin\left(\frac{\pi}{10}t - \frac{\pi}{2}\right) + 17$ meters. Find the first time $t > 0$ (minutes) when the rider is at height 17 m and rising.